

REVIEW

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Bibliometric overview of global electrocatalytic water treatment research

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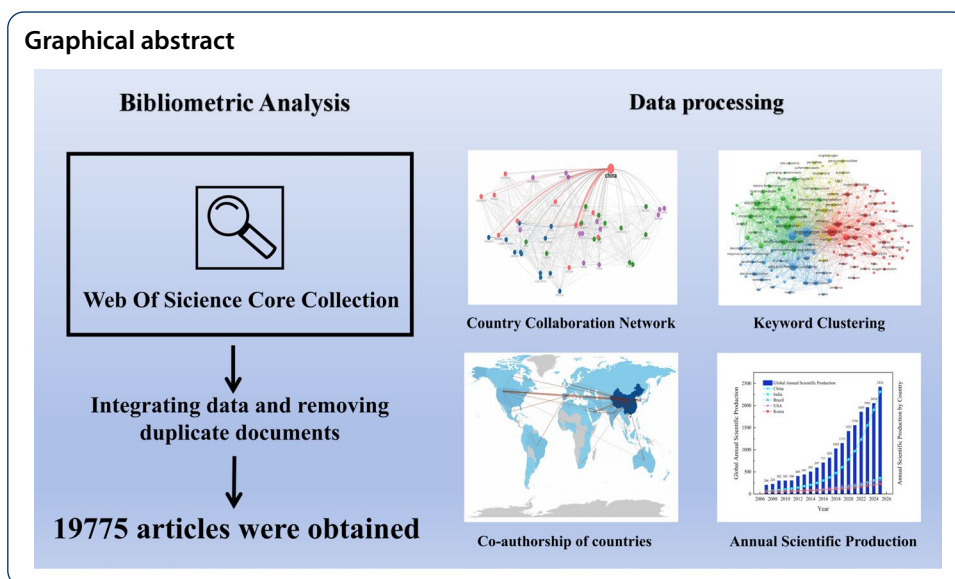
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Abstract

Remediation of refractory organic pollutants in aquatic environments has become a core research priority in global environmental science, and electrocatalytic oxidation has emerged as one of the most promising advanced oxidation processes owing to its high degradation efficiency, environmental compatibility and operational flexibility. Herein, a bibliometric methodology is adopted to systematically elucidate the global research landscape, intellectual structure and evolutionary dynamics of this field: based on 19,775 relevant publications retrieved from the Web of Science Core Collection covering 2007 to 2025, a comprehensive quantitative analysis is performed using the visualization tools VOSviewer and Biblioshiny, covering publication characteristics, subject distribution, national and institutional collaboration networks, keyword co-occurrence clustering, temporal evolution of research hotspots, and profiles of highly cited papers. The results indicate that the overall research output maintains a sustained growth trend and has entered a phase of rapid expansion since 2018; The Chinese Academy of Sciences holds the highest centrality in the institutional collaboration network, acting as a major regional cooperation hub within Asian academic networks. Research themes can be grouped into four major clusters: development of catalytic materials, reaction kinetics and mechanism interpretation, pollutant mineralization performance, and electro-Fenton systems, while boron-doped diamond electrodes, carbon-based materials and nano-modification strategies constitute the mainstream directions of current material research. The global collaboration network presents a multi-center clustered pattern, with European and North American countries showing a higher ratio of international co-authorship, and China presents the highest total publication output, while European and American nations exhibit wider international cooperation breadth and higher average citation per article. This study systematically maps the development trajectory and intellectual framework of electrocatalytic water treatment research, and identifies that low-cost catalytic material development, multi-technology coupled process optimization, by-product control and intelligent process optimization will be the core future research directions, providing objective data support and decision-making references for research planning, academic collaboration and technology deployment in this field.

Keywords Electrocatalytic oxidation, Water treatment, Bibliometric analysis, Research collaboration networks, Knowledge mapping





1 Introduction

Rapid industrialization, agricultural development, and urbanization have introduced a wide range of complex chemical pollutants into freshwater systems [1]. Industrial wastewater is discharged without proper treatment, which produces persistent organic pollutants (POPs), chemicals, endocrine disrupting chemicals and emerging contaminants into aquatic habitats. These substances resist normal biological treatment processes due to their structure, chemical stability, poor biodegradability, with a half-life ranging from months to decades in nature. Extensive accumulation of these pollutants causes cascading ecological effects such as aquatic toxicity, bioaccumulation through food chains, and long-term threats to human health through contaminated drinking water [2].

Conventional water treatment technologies such as coagulation, sedimentation, biological degradation, and membrane filtration are highly effective for removing bulk pollutants, suspended solids, and readily biodegradable organics. However, they show limited removal efficiency for many refractory persistent organic pollutants. For example, activated sludge processes typically achieve less than 30% removal for many synthetic pharmaceuticals and pesticides due to their poor biodegradability. Membrane technologies can effectively retain pollutants via size exclusion, but they only separate and concentrate contaminants rather than degrade them, requiring additional treatment for concentrated streams [3, 4]. This technology has spurred the exploration of Advanced oxidation processes (AOPs), which have been developed to degrade these recalcitrant pollutants, aiming to convert them into carbon dioxide, water, and inorganic ions through oxidative decomposition. Electrocatalytic oxidation has been a popular choice as it offers remarkable degradation efficiency, flexibility and environment compliance advantages [5, 6].

Electrocatalysis uses electrochemical reactions at engineered electrode surfaces to decompose organic pollutants in two complementary ways: direct and indirect oxidation. Direct oxidation occurs when pollutants adsorb onto the anode surface and are degraded via heterogeneous electron transfer at the electrode-solution interface, involving dehydrogenation, bond cleavage, and partial mineralization. This pathway is

dominant for pollutants with strong electrode affinity, although the overall oxidation rate is often relatively limited. Degradation is mediated by electrochemically generated reactive species that diffuse into the bulk solution, including: (i) reactive oxygen species such as hydroxyl radicals ($\cdot\text{OH}$), hydrogen peroxide (H_2O_2), and sulfate radicals ($\text{SO}_4^{\cdot-}$); (ii) reactive chlorine species such as Cl_2 , hypochlorous acid (HOCl), and chlorine dioxide (ClO_2), generated from chloride oxidation at the anode. Indirect oxidation achieves non-selective degradation of pollutants in the bulk solution [7–9].

Compared to other AOPs, electrocatalysis offers both (i) low temperature and pressure, (ii) minimal chemical reagent consumption apart from electrolyte salts, (iii) scalable modular design suited for different treatment capacities, (iv) renewable energy for sustainable operation and (v) potential resource recovery via selective oxidation or electrochemical synthesis, but still challenges remain including high energy consumption, electrode fouling reduction, and control of toxic chlorinated byproducts in chloride matrices [10–12].

Bibliometrics quantitatively analyzes publication output, citation patterns, and collaboration networks, and transforms large-scale literature datasets into knowledge maps reflecting research progress, intellectual structure, and evolutionary dynamics, providing evidence-based support for research management [13]. Statistics and mathematics are used in this method for the study of scholarly publications and citation patterns and collaboration networks in a research domain. Academic publications and their citation networks are analyzed for the investigation of numerical patterns and searching for open links in the literature [14]. Researchers use several bibliometric indicators that meet multiple analysis goals: journal impact in specific research directions, research trends, prominent researchers and authority institutions, international collaboration patterns, research challenges and promising future research directions [15]. Bibliometric visualization tools such as VOSviewer, CiteSpace and Biblioshiny have been developed. Researchers use bibliometric methods to obtain different analysis goals. Key metrics are journal impact in certain areas, research trends in some areas, prominent research institutes and top institutes of science, global cooperation, current research trends, opportunities for future work [16]. These software programs provide multidimensional visualization of bibliometric results by turning complex datasets into interpretable graphics. These tools enable efficient knowledge extraction and systematic trend matching from large information sets. Co-citation networks are introduced that show theoretical foundations, co-authorship maps which show collaboration patterns, and keyword co-occurrence clusters that represent themes.

Compared to existing bibliometric studies on two categories of electrochemical advanced oxidation processes, this study demonstrates significant incremental value across multiple dimensions [17, 18]. While Macias-Quiroga et al. 's analysis generalized findings across all advanced oxidation processes and Brdaric et al. 's research covered only the past five years, our study specifically focuses on electrocatalytic water treatment as a specialized subfield, extending the time span to 2007–2025 and incorporating nearly 20,000 relevant publications—a sample size orders of magnitude larger than both previous studies. This approach not only avoids the superficial coverage inherent in general AOP analyses but also overcomes the limitations of short-term studies, enabling a comprehensive depiction of the field's long-term evolution from inception to rapid expansion, thereby enhancing the generalizability and timeliness of conclusions. Regarding

collaboration network analysis, while prior studies primarily relied on visualization and basic publication/citation statistics, our study introduces three key network metrics—betweenness, closeness, and PageRank—to quantitatively assess the pivotal roles and information transmission efficiency of leading countries/institutions within collaborative networks. We further standardize analyses by distinguishing single-country publications from multi-country collaborations, calculate cooperation ratios, and systematically compare regional research autonomy and international collaboration depth, providing robust quantitative support for interpreting global collaboration patterns. In terms of knowledge structure analysis, this study employs a dual-threshold keyword co-occurrence network combined with hierarchical word cloud visualization, clearly distinguishing between the core knowledge framework of the field and peripheral sub-topics. Compared to single-dimensional keyword networks in previous studies, it provides a more refined depiction of thematic hierarchies and clearer logical organization. Regarding high-citation literature analysis, the study transcends mere content replication by correlating highly cited works with thematic clusters, highlighting the theoretical underpinnings of classical literature for various research domains. By examining generational shifts in high-citation literature, it identifies patterns of paradigm evolution in field research, enabling quantitative analysis to effectively inform understanding of disciplinary development trajectories. In terms of future directions, distinct from existing studies that offer broad, generic recommendations, this research proposes four targeted strategies—low-cost long-life electrode development, multi-technology integration processes, chloride by-product control, and intelligent process optimization—all supported by concrete data and practical applicability. Overall, this study addresses both the lack of focus on electrocatalysis in general AOP-based quantitative research and the absence of long-term evolutionary analysis in short-cycle EAOP studies, delivering a comprehensive, systematic, and actionable framework for electrocatalytic water treatment research.

2 Methodology

2.1 Data collection

Web of Science database was used to systematically cover global research in multiple fields. It was used for an analysis of electrocatalytic degradation of pollutants. We used the search query TS (Topic Search) = (Electrocatalysis OR Electrocatalytic oxidation OR Electrochemical oxidation OR electro-Fenton OR Electrochemical advanced oxidation process OR EAOP OR Anodic oxidation OR Electrocatalytic degradation) AND TS = (Pollutants OR Contaminants OR Wastewater OR Micropollutants OR Water treatment OR Wastewater treatment) to appropriately characterise the study. The search has focused on Web of Science Core Collection, SCI-EXPANDED and SSCI indices covering 2007–2025 publications. Our last extraction was conducted June 19, 2026. We selected 19,775 publications which we first categorized and analyzed.

A statistical analysis of publication patterns in electrocatalytic studies of pollutants removal shows different trends, Research articles account for 90.10% of all publications as they represent original experimental and theoretical findings and technological advances and review articles account only for 9.90% of publications [19]. These reviews play a vital academic role by systematically synthesizing research progress, consolidating existing knowledge, and providing foundational insights particularly valuable for new investigators, while also charting potential directions for future research.

2.2 Analysis method

Social Network Analysis (SNA) probes how people and groups connect by mapping and analyzing relationships. For SNA nodes represent actors (e.g. single researchers, institutions, or even countries), edges represent links or contacts between them, and thus the social structure can be represented, connection patterns, and relationships strength and closeness.

SNA has many applications for mapping international collaborations and global cooperation, such as journal citation networks, Co-occurring keywords and study the evolution of research networks. In doing so, researchers learn from challenging data²⁰. It is used by a specialist such as VOSviewer and Biblioshiny. VOSviewer is used for network visualization, while Biblioshiny supports in-depth bibliometric analysis and presents results in intuitive visual forms.

VOSviewer can be used by researchers to build and visualize complex networks of links between authors, institutions, countries, and publications. Several techniques can be tested using this software, such as Co-citation analysis, term co-occurrence analysis, and bibliographic coupling, which uncover research literature structure, small clusters, and study development [20]. In this work, term co-occurrences analysis was used to find key topics, key concepts and map the field main structure, providing information on research fronts.

Biblioshiny is a graphical user interface application based on R and the bibliometrix package, which enables researchers to perform advanced bibliometric analysis without extensive programming skills [21]. Researchers can use it to carry out studies on large data sets, to analyze several papers, and to find meaningful patterns in their research [22]. Biblioshiny's keyword functions were used to examine co-occurrence networks in depth, pointing out core research themes, current research priorities, discovering key linkages between ideas, tracking the development of topics, pointing to future directions, and providing a multidimensional vision for future work.

3 Results and discussion

3.1 Document types and numbers

This study investigated electrocatalytic technology for pollutant treatment and analyzed how researchers were receiving papers in different research categories. We extracted relevant papers from the Web of Science Core Collection database and grouped them into categories. Each publication was classified according to the first research category selected in Web of Science subject classification and we calculated the distribution of publications in categories. The distribution revealed different research topics with some being large and others being emerging or special research domains. Research categories with less than 2% of total publications were not included in the figure labels. The full distribution of papers in research categories is given in Fig. 1.

There are 7 main research areas for which each of them make over 5% of publications. Engineering Chemical 14.5%, Engineering Environmental 13.8%, Environmental Sciences 13.5%, Chemistry Physical 8.6%, Electrochemistry 6.6%, Chemistry Multidisciplinary 6.2%, Materials Science Multidisciplinary 6.2%. These seven areas are among the major sources for electrocatalytic pollution research. Engineering is one of the most well-represented areas leading to solutions to pollution problems. Basic and applied research are closely related [23–25]. Physical chemistry offers fundamental insights [26,

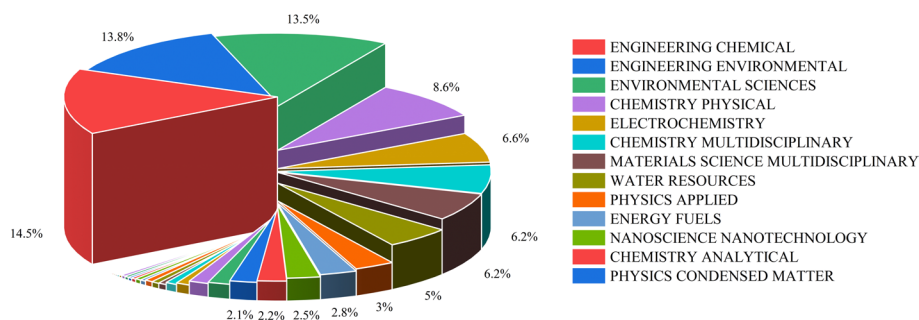


Fig. 1 Web of science categories

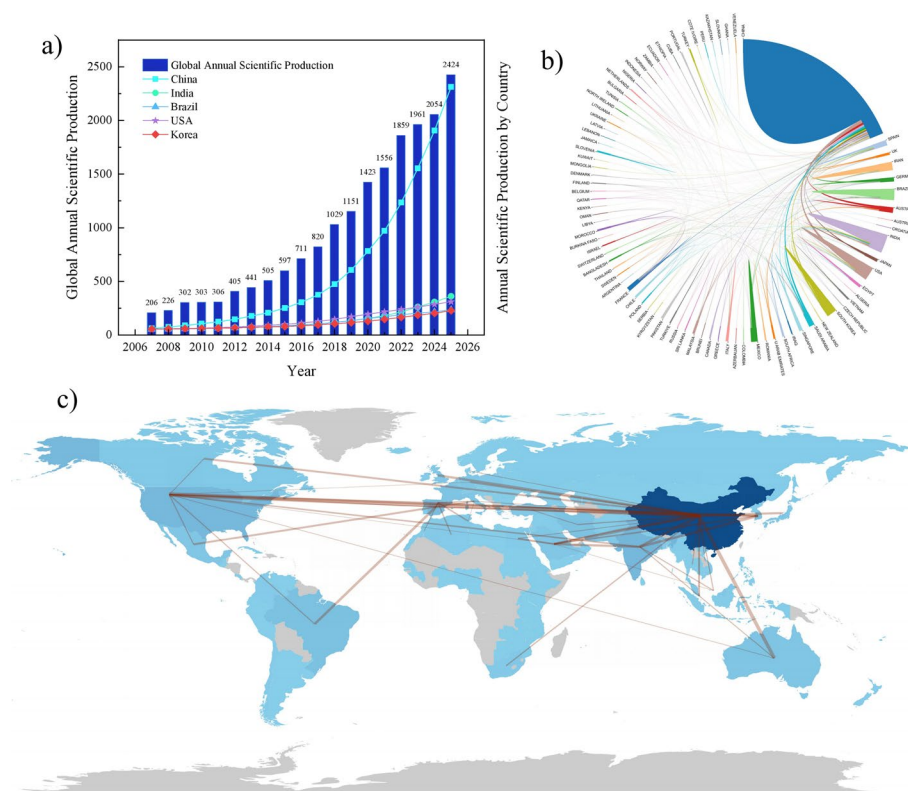


Fig. 2 a. Annual scientific production b. Corresponding author's countries c. Corresponding author's countries' scientific production

27], while Environmental sciences emphasize practical applications [28, 29], and multi-disciplinary chemistry fosters collaboration across fields, creating valuable synergies [30, 31]. Materials science advances the field through novel materials development, enabling improved treatment methods and supporting technological innovation [32, 33]. Electrochemistry provides an essential understanding of reaction mechanisms, collectively driving continuous progress, and expanding the boundaries of innovation [34–36].

3.2 Current publication trends and active countries

Annual scientific production output serves as a key indicator of research activity within a field. As shown in Fig. 2, there were only a few publications until 2017, but since 2018,

scientists produced steadily. Five countries contributed the majority of the published works, with China ranking first, followed by India, the United States, Korea and Brazil.

China initiated relevant research at a relatively early stage, and the development of the field entered a phase of rapid acceleration after 2018, with a significant increase in annual research output. Chinese scholars have carried out extensive international cooperative research. Other major countries including India and Brazil have also carried out active exploration in this research area: India's development rate has remained at a relatively low level, while Brazil has maintained a steady growth trend. However, both the growth rate of research and development and the overall contribution of the above-mentioned countries are lower than the impressive development momentum achieved by China. On the global scale, this research field is in a continuous growth trend, with China having accumulated abundant domestic research outputs, while European and North American teams maintain stronger foundational theoretical influence. The diversified participation pattern formed by this global layout has brought more diverse research perspectives to the development of the field from the international academic community.

China's research area is the development of new catalysts [37], reaction mechanisms [38], and electrocatalytic technology for pollutants treatment [39]. These studies have led to theoretical results in electrocatalysis, as well as environmental remediation efforts. However, other countries tend to take other topics, where India focuses on specific pollutant types and Brazil focuses on reaction conditions. Although these are more focused, they provide useful insights and are more diverse in the field. Therefore, global research presents complementary characteristics. China produces a broad spectrum of material-oriented experimental outputs, while other countries carry out targeted niche research, and such diversified research perspectives jointly promote the development of this field.

3.3 Keyword co-occurrence clustering and temporal evolution of research hotspots

This section integrates two complementary bibliometric analyses to avoid redundant discussion. Static keyword co-occurrence clustering reveals the fixed static knowledge architecture of this field, while temporal cumulative keyword trends track dynamic shifts of research hotspots over two decades. The two dimensions jointly deliver static-structural and dynamic-evolution perspectives.

As shown in Fig. 3, VOSviewer software was used to analyze publication data to examine keyword relationships and map research domains. The resulting visualizations present organized clusters, each distinguished by color. When the co-occurrences threshold is 100, we observed large networks of connected terms showing the diversity and depth of the field. When the threshold is raised to 200, only highly correlated core terms remain, forming a streamlined knowledge skeleton with larger nodes and thicker connecting lines. The position of nodes reflects their structural role: centrally located nodes have more extensive connections and act as key hubs in the knowledge network, while peripheral nodes represent niche or emerging subtopics.

Co-occurrence clustering analysis is a well-established bibliometric technique that examines the co-occurrence relationships among keywords within scientific literature. This method identifies often-occurring keyword pairs and groups close-related words into groups that map the correlations and groups from a number of research topics. This enables researchers to quickly capture research hotspots, themes and ties. Using

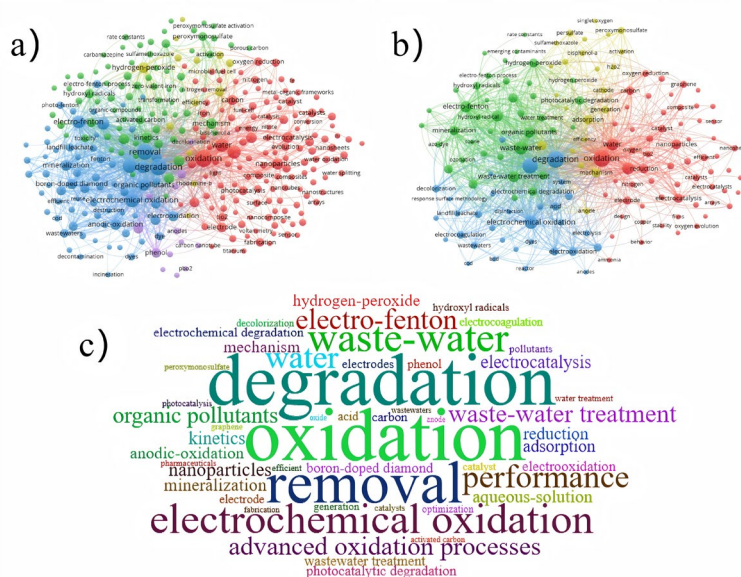


Fig. 3 **a.** The minimum co-occurrence frequency is 100. **b.** The minimum co-occurrence frequency is 200. **c.** Word cloud for frequently used keywords

this method, researchers are able to identify convergence points and branching trends in research directions.

The word cloud visualizes keyword frequency via font size, enabling researchers to quickly identify dominant research themes and detect potential research gaps by comparing core and peripheral terms. The keywords in addition to the keyword feature highlight gaps or missing areas.

Figure 3 presents the keyword co-occurrence networks and word cloud generated by VOSviewer, which collectively delineate the intellectual structure, thematic clusters and hotspot distribution in the field of electrocatalytic water treatment. Panel (a) illustrates the co-occurrence network with a minimum co-occurrence threshold of 100, which covers a comprehensive set of keywords to present the full thematic spectrum of the field; in this network, node size represents the occurrence frequency of each keyword, line thickness reflects the co-occurrence strength between terms, and three distinct color-coded research clusters can be identified: the blue cluster centered on degradation, oxidation and electrochemical oxidation gathers terms including boron-doped diamond, electrode, wastewater and phenol, corresponding to the research domain of anodic oxidation processes and electrode materials; the red cluster dominated by catalyst and nanoparticles incorporates graphene, composite, metal-organic frameworks and oxygen reduction, representing the development of nanocatalysts and carbon-based functional materials; the green cluster featured by electro-Fenton and hydrogen peroxide covers hydroxyl radicals, mineralization and kinetics, reflecting research focus on indirect oxidation pathways and reaction mechanisms, while peripheral terms such as persulfate, peroxymonosulfate, landfill leachate and dyes further indicate the diversified expansion of the field into specialized sub-directions. When the co-occurrence threshold is increased to 200 in panel (b), only highly correlated core keywords remain, streamlining the network into a clear core knowledge skeleton: degradation and oxidation serve as the absolute hub nodes connecting all thematic clusters, and the three core clusters

maintain close interconnection, which verifies that the research mainline of “material development–process construction–mechanism interpretation–performance evaluation” is widely recognized and structurally stable in this field. Panel (c) displays the word cloud of high-frequency keywords where font size directly corresponds to occurrence frequency, and the terms can be categorized into three hierarchical layers: the largest terms including degradation, oxidation, electrochemical oxidation, removal and wastewater form the foundational conceptual layer that defines the core research scope; terms such as electrocatalysis, organic pollutants, kinetics, mineralization, nanoparticles, boron-doped diamond, hydrogen-peroxide, electro-Fenton and performance constitute the core hotspot layer, covering four critical dimensions of treatment targets, key materials, core processes and evaluation indicators; supporting terms like phenol, carbon, catalyst and advanced oxidation processes act as the subdivided supporting layer corresponding to specific research approaches. Overall, this set of figures demonstrates that electrocatalytic water treatment research has developed a stable and well-defined intellectual structure with a highly integrated research system revolving around electrode materials, oxidation processes, reaction mechanisms and degradation performance, while core research hotspots remain focused on high-efficiency electrode development, electro-Fenton system optimization and pollutant mineralization enhancement, with continuous extension into emerging subfields such as persulfate activation and emerging contaminant treatment. Notedly, the three visually defined clusters can be further subdivided into four thematic domains by research content: catalytic material development, reaction kinetics and mechanism interpretation, pollutant mineralization performance, and electro-Fenton systems, which is consistent with the overall thematic framework of the field.

Keywords were also found with a random distribution and their relationships. The system calculated the occurrence frequency of each keyword, excluded commonly cited terms such as “electrocatalysis”, “pollutants”, and ubiquitous words like “water” during searches, and identified the top 10 most frequently occurring terms. As shown in Fig. 4, these high-frequency terms reflect the primary research directions and future focus areas. This systematic organization approach not only facilitates further analyses, but also effectively explores intrinsic conceptual relationships, thereby providing a reliable research foundation.

The top 10 keywords are listed in descending order: Performance (Frequency = 2284), Organic pollutants (Frequency = 1594), Nanoparticles (Frequency = 1339), Kinetics (Frequency = 1264), Mineralization (Frequency = 1231), Hydrogen-peroxide (Frequency = 1158), Boron-doped diamond (Frequency = 991), Carbon (Frequency = 908), Phenol (Frequency = 875), Catalyst (Frequency = 752). This keyword frequency data suggests a research community strongly focused on electrocatalytic treatment of pollutants. Specifically, the dominance of “Performance” and “Kinetics” underscores that quantitative evaluation of degradation efficiency and mechanistic interpretation of reaction pathways remain the central pursuits in this field. Researchers consistently prioritize establishing kinetic models (typically pseudo-first-order kinetics) to quantify pollutant decay rates, and linking kinetic parameters to electrode properties and operational parameters for systematic process optimization.

The high occurrence of “Organic pollutants”, “Phenol” and “Mineralization” reflects two core characteristics of current research: on one hand, phenolic compounds serve

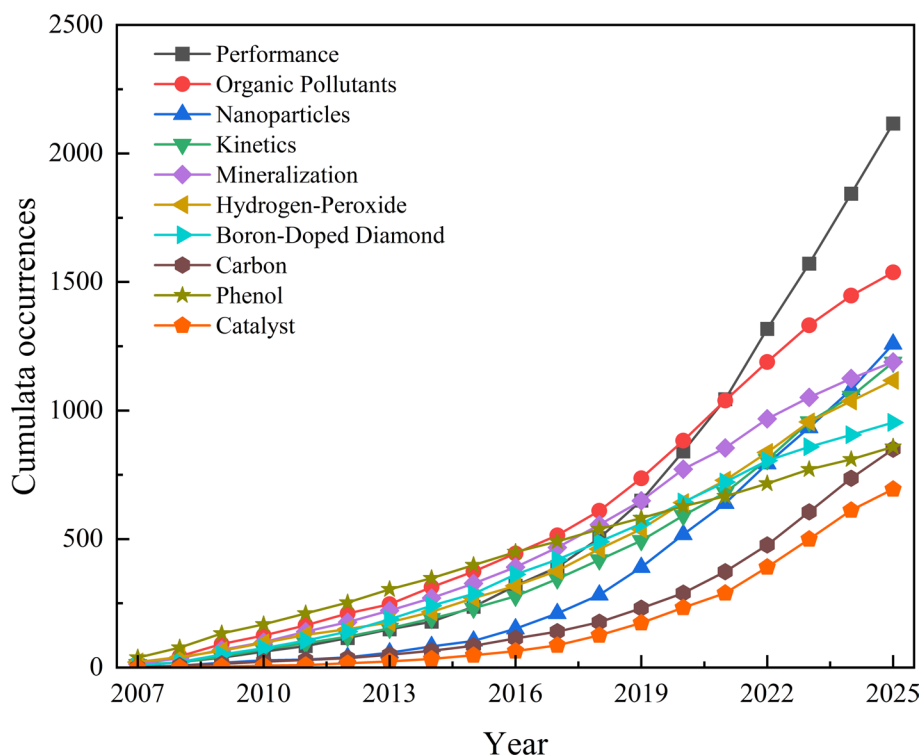


Fig. 4 Words' frequency over time

as the most widely adopted model pollutants for verifying electrocatalytic oxidation performance, owing to their representative refractory aromatic structure and universality in industrial wastewater [40, 41]; on the other hand, the research goal has evolved from simple decolorization or parent compound removal to complete mineralization of organic matter into CO_2 and inorganic ions, as evidenced by the widespread adoption of total organic carbon (TOC) removal rate as a rigorous performance metric [42, 43].

In terms of catalytic material systems, the frequent appearance of “Nanoparticles”, “Carbon”, “Boron-doped diamond” and “Catalyst” outlines the mainstream development trajectory of electrocatalytic materials [44, 45]. Boron-doped diamond, as a classic high-efficiency anode material with an ultra-wide potential window, weak adsorption capacity and excellent corrosion resistance, maintains enduring research attention in electrochemical advanced oxidation processes; carbon-based materials serve as versatile electrode substrates or cathode candidates with tunable pore structure, high conductivity and low cost; meanwhile, nanoparticle modification and heterogeneous catalyst design have become dominant modification strategies to enhance electrocatalytic activity, accelerate interfacial charge transfer and boost the generation of reactive oxygen species [46–50].

Notably, the high frequency of “Hydrogen-peroxide” indicates the prominent position of electro-Fenton and in-situ H_2O_2 generation systems in the field [51, 52]. The cathodic reduction of dissolved oxygen to produce H_2O_2 , coupled with catalytic activation to generate strong oxidizing hydroxyl radicals, constitutes a critical advanced oxidation pathway for the degradation of refractory organic pollutants, and has become a key branch parallel to anodic direct oxidation [53, 54].

Beyond individual frequency statistics, the inherent associations among these keywords delineate distinct yet interconnected research clusters: material innovation drives the improvement of degradation performance, kinetic analysis provides quantitative support for mechanism elucidation, and mineralization efficiency serves as the ultimate evaluation criterion for pollutant treatment systems [55, 56]. These interwoven conceptual relationships collectively map the intellectual structure of electrocatalytic wastewater treatment research, highlighting the close integration of material science, reaction engineering and environmental application [57, 58].

Such keyword-based bibliometric analysis not only intuitively presents the current research landscape and core hotspots, but also provides a quantitative foundation for identifying under-explored directions and emerging research frontiers. It lays a solid data basis for subsequent semantic analysis, topic evolution tracking and co-occurrence network construction, helping researchers quickly grasp the overall development context of the field and locate potential breakthrough points.

3.4 Citation trends

Top papers are often cited as highly influential. They can introduce new theories or give important experimental data that will inform subsequent research and can be used as a reference for other scholars. Citation analysis is used to assess research quality and impact.

Table 1 lists the top 8 most cited publications in the dataset, with metrics including total citations (TC), TC per year, and normalized TC. The included papers are characterized in terms of their research topics, methodological approaches and experimental procedures. The listed studies feature innovative experimental designs, draw well-founded conclusions with generalized applicability, and frequently explore under-researched domains, propose novel methodological approaches, construct new testing frameworks, and expand the boundaries of the research field. Collectively, these high-impact studies advance the accumulation of disciplinary knowledge, provide both theoretical foundations and practical guidance for subsequent research, and ultimately promote the development of scientific progress.

In the construction of knowledge structure, the collected literature systematically covers four core research domains within the field of electrocatalytic water treatment: (i) selection and modification of catalytic materials, (with research emphasis placed on the development of non-ferrous metal catalysts and metal-free catalysts; (ii) design and parameter optimization of electrochemical reactors, covering key influencing variables including anode material, current density and pH; (iii) comparative analysis of energy efficiency across different advanced oxidation process (AOP) technologies, which acts as

Table 1 Most global cited documents

Rank	Paper	Total citations	TC per year	Normalized TC
1	Brillas E, 2009, Chem Rev [59]	3022	167.89	30.97
2	Wen JQ, 2017, Appl Surf Sci [60]	2748	274.80	45.17
3	Miklos DB, 2018, Water Res [61]	2367	263.00	35.82
4	Martínez-Huitle CA, 2009, Appl Catal B-Environ [62]	2174	120.78	22.28
5	Bokare AD, 2014, J Hazard Mater [63]	2099	161.46	29.35
6	Panizza M, 2009, Chem Rev [64]	2071	115.06	21.22
7	Moreira FC, 2017, Appl Catal B-Environ [65]	1883	188.30	30.95
8	Oturan MA, 2014, Crit Rev Env Sci Tec [66]	1830	140.77	25.59

a quantitative benchmark; (iv) process integration and engineering application, such as the combined Photo-electro-Fenton process. These four domains are highly consistent with the high-frequency clustering results obtained from keyword co-occurrence network analysis in bibliometrics, and provide an important theoretical basis for the topic identification and evolution analysis conducted in this study.

As summarized in Table 1, only one out of the eight top-cited landmark review papers is led by Chinese corresponding authors. This quantitative evidence confirms that sheer publication volume cannot be equated with foundational theoretical influence or internationally recognized landmark outputs. Most high-impact classic reviews were published by European research groups between 2009 and 2017, systematically establishing the fundamental framework of electro-Fenton and anodic oxidation mechanisms. In contrast, most Chinese publications focus on applied material modification at lab scale, with shorter citation accumulation cycles and lower cross-field recognition.

Furthermore, the existing literature collectively reveals the temporal evolution pattern of research priorities in this field: the research focus has shifted from early-stage exploration of single pollutant degradation and material screening to multi-process comprehensive comparison, actual wastewater treatment verification, and integrated assessment of energy efficiency and by-product risk. This evolution explicitly reflects the paradigm shift in the field from a “laboratory performance-oriented” research approach to an “engineering feasibility and sustainability-oriented” research strategy. Meanwhile, the common bottlenecks that have been consistently identified in existing research—including the dependence of iron-based Fenton processes on acidic pH conditions, the service life and cost issues of BDD electrodes, the generation of chlorinated by-products during the treatment of chlorine-containing wastewater, and the insufficient utilization of visible light—remain active research topics in bibliometric analysis. This trend indicates that future research breakthroughs will focus on multi-technology integration, development of low-cost catalytic materials and intelligent process optimization, which provides a clear reference framework for the subsequent term detection and frontier identification work in this study. These research challenges are frequently reflected in bibliometric data through keywords such as “heterogeneous catalysis”, “solar-driven systems” and “by-product control”, which suggests that multi-technology coupling (e.g., photoelectrochemical Fenton systems coupled with membrane separation), development of low-cost catalysts (e.g., biomass-derived carbon materials) and intelligent process optimization (e.g., machine learning-assisted parameter optimization) will become core research directions in this field. Accordingly, the bibliometric analysis presented in this paper will further focus on “identifying emerging research frontiers through interdisciplinary integration (e.g., catalytic materials + energy efficiency + process coupling)”, so as to provide more forward-looking quantitative insights for the development of the field.

3.5 Keywords usage by countries and references

The Three-Field Plot effectively illustrates the global distribution of highly cited publications, visually mapping research contributions by country, institution, author. These charts can find top countries, top institutions, and top authors and also reveal collaboration relationships between them. From keyword patterns, the chart can guide research relationships, national strengths, institutions, as well as authors’ influence (Fig. 5).

Table 2 Betweenness and closeness of each institution

Rank	Institution	Betweenness	Closeness	PageRank
1	Chinese Academy Of Sciences	236.868	0.018	0.132
2	University Of Chinese Academy Of Sciences, CAS	11.569	0.015	0.059
3	Research Center For Eco-Environmental Sciences (Rcees), CAS	2.004	0.013	0.04
4	University Of Science And Technology Of China, CAS	1.688	0.013	0.028
5	Harbin Institute Of Technology	58.42	0.016	0.038
6	Nankai University	66.21	0.016	0.03
7	Tianjin University	4.912	0.014	0.024
8	Tsinghua University	24.028	0.016	0.029
9	Hunan University	7.19	0.014	0.02
10	Dalian University Of Technology	3.005	0.013	0.016
11	Sichuan University	2.634	0.014	0.013
12	Zhejiang University	11.045	0.014	0.015
13	Tiangong University	0.206	0.011	0.009
14	South China University Of Technology	1.616	0.013	0.012
15	Zhejiang University Of Technology	1.228	0.012	0.01
16	Huazhong University Of Science And Technology	2.154	0.012	0.009
17	Nanjing University	5.612	0.014	0.015
18	Central South University	1.938	0.013	0.013
19	Xi'An University Of Architecture And Technology	1.723	0.012	0.011
20	China University Of Geosciences	3.966	0.013	0.013
21	China University Of Mining And Technology	0.248	0.011	0.01
22	Jilin University	1.338	0.013	0.009
23	Wuhan University	2.642	0.012	0.011
24	Beijing Normal University	6.654	0.013	0.013
25	Shandong University	11.609	0.014	0.013

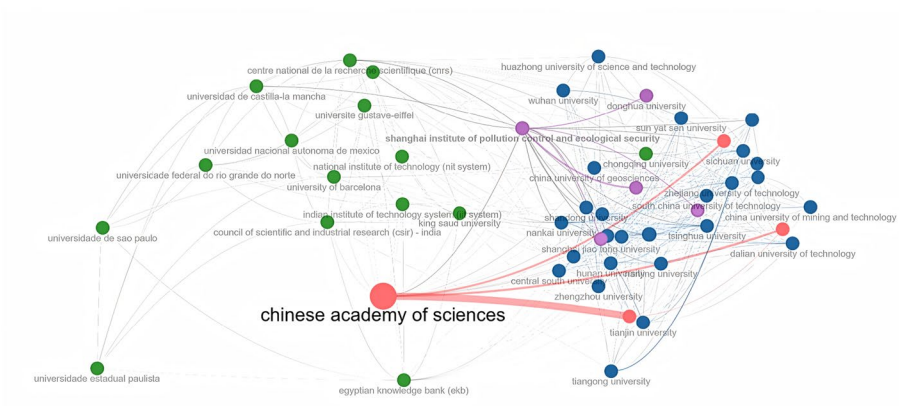


Fig. 6 Institution collaboration network

Figure 6 shows that there are well-established patterns of institutional collaboration. There are a large number of organizations, which make many connections with partners, and a few other ones are still far away. The Chinese Academy of Sciences is one of the biggest institutions as the betweenness center. Others work with it, and their contributions advance the development of the field.

There are also universities with different interests in the collaboration chain such as Nankai University and Tianjin University. Their contributions are complementary to each other. Special research institutes dedicated to specific area and coordinate with each other to solve technical challenges, and they all contribute to an unbalanced effort.

3.7 Country collaboration network analysis

Bibliometrix analysis is instrumental in assessing national influence and clarifying the global distribution of research strength. Table 3 presents data for the top 25 countries, including publication counts, single country publications (SCP), multiple country publications (MCP), and multiple country publications rate (MCP Ratio). This summarizes each country's research and level of cooperation, identifying the leading countries and emphasizing the importance of international partnerships.

Table 3 indicates that China ranks first in terms of publication volume, followed by India and Spain. China published a significantly larger number than other countries, showing that it is one of the most important research fields in electrocatalytic field for pollutant treatment. Their work is very wide, from fundamental research to real applications. China has the largest total publication quantity, without equivalent advantages in international cooperation and high-impact landmark papers. Other developing countries continue to contribute, and their work makes the field continuously active.

The MCP ratio acts as a normalized metric to evaluate international collaboration depth. China's MCP ratio is only 17.3%, far lower than France (59.8%), Australia (55.8%), UK (54%) and USA (31.3%). This gap indicates most Chinese outputs are independent domestic studies, while Western research relies heavily on cross-border joint projects to generate high-impact collaborative papers.

China exhibits prominent performance in both domestic and international academic publications, with its SCP and MCP both ranking first globally. This finding can be primarily attributed to China's strong domestic research capacity and high level of international academic recognition. India, Spain, the United States, and Brazil also maintain

Table 3 Number of publications and partnerships by country

Rank	Country	Articles	SCP	MCP	MCP ratio/%
1	China	9512	7871	1641	17.3
2	India	1232	947	285	23.1
3	Spain	797	554	243	30.5
4	USA	760	522	238	31.3
5	Brazil	754	519	235	31.2
6	Korea	636	430	206	32.4
7	Iran	627	497	130	20.7
8	Mexico	406	261	145	35.7
9	France	373	150	223	59.8
10	Italy	330	200	130	39.4
11	Canada	270	192	78	28.9
12	Turkey	259	231	28	10.8
13	Australia	240	106	134	55.8
14	Germany	231	137	94	40.7
15	Japan	195	154	41	21
16	Poland	163	131	32	19.6
17	Malaysia	162	85	77	47.5
18	Egypt	158	87	71	44.9
19	Portugal	144	85	59	41
20	United Kingdom	139	64	75	54
21	Saudi Arabia	122	54	68	55.7
22	Pakistan	113	35	78	69
23	Tunisia	113	67	46	40.7
24	South Africa	105	72	33	31.4
25	Algeria	98	50	48	49

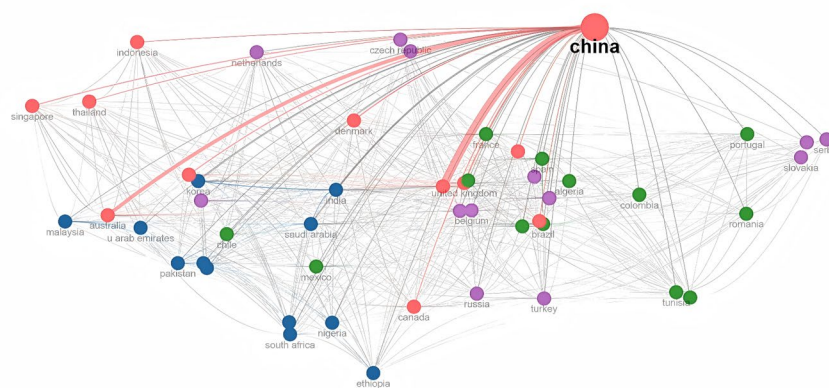


Fig. 7 Country collaboration network

high SCP and MCP values, and their overall scientific contributions are second only to China. These countries have established relatively stable and mature research teams in this research field, and possess top-tier academic influence globally. Although the SCP and MCP values of the other listed countries exhibit a gradual decreasing trend, all of these countries have developed characteristic research directions in the corresponding field, providing diversified support for the development of this discipline.

International cooperation networks form four distinct clusters. The red cluster is led by China; the blue cluster centers on India and South Africa; the purple cluster is dominated by Brazil; the green cluster is represented by Spain and other European countries. Each cluster forms close internal knowledge sharing and collaborative patterns. The four colors denote different cooperative modes, each of which realizes knowledge sharing within the community. The visualization result intuitively presents the scale and diversity of international scientific research cooperation.

From 2007 to 2025, numerous countries worldwide actively engaged in and significantly advanced research on electrocatalytic pollutant removal technology—a cutting-edge field within environmental science and technology. The visualization of the international collaboration network shown in Fig. 7 clearly demonstrates a marked enhancement in cooperative relationships among nations, leading to closer and more frequent collaborations as well as the establishment of specialized research teams with unique collaborative frameworks and knowledge-sharing mechanisms.

Notably, by 2025, international cooperation had become the norm in this field: countries globally joined this expanding research community, joint projects continued to grow in both scale and number, the depth and breadth of international collaboration improved substantially, and academic ties became increasingly robust and diverse. Meanwhile, emerging participants and research institutions further enriched the global collaboration network, fostering cross-border knowledge exchange and collaborative innovation.

The United States and Spain have high international collaboration rates and strong citation impact, focusing on fundamental mechanisms and high-performance electrodes. India and Brazil focus on specific industrial pollutants and low-cost materials. Each country has formed its own research characteristics, jointly promoting diversified development of the field.

3.8 Limitations

This study is solely based on the Web of Science Core Collection, and does not include literature from other databases such as Scopus, CNKI and regional non-English journals. This may lead to omission of some regional research outputs and non-English publications, resulting in a certain degree of English-language and international journal bias.

The search is conducted with predefined keyword combinations. Some interdisciplinary studies with unconventional terminology or newly emerging research branches may not be fully covered, which brings inherent retrieval deviation to the dataset.

The analysis is mainly based on quantitative indicators such as publication count, citation frequency and keyword co-occurrence, which cannot directly evaluate the academic quality and innovation depth of individual studies. In addition, citation data has a natural time accumulation effect, and high-quality papers published in recent years may be underestimated in citation-based evaluation.

Cooperation networks are constructed only based on author affiliation information, which cannot capture informal academic exchanges and implicit collaborative relationships. Keyword clustering is also affected by authors' indexing habits, and there may be deviations in thematic boundary division.

4 Conclusions

This study conducts a systematic bibliometric investigation of electrocatalytic technology for water pollutant treatment, based on 19,775 publications from the Web of Science Core Collection covering 2007–2025. Through multi-dimensional analysis of document types, annual publication trends, keyword evolution, citation patterns, and collaboration networks at both national and institutional levels, we delineate the global research landscape, intellectual structure and evolutionary dynamics of the field, and identify the underlying linkages and driving forces that shape its development.

Our findings confirm a continuous and accelerating growth in global research output, with a marked expansion phase starting in 2018. China has the largest publication volume across all nations, yet its international collaboration ratio and proportion of top highly cited papers lag behind European and North American research communities, while the Chinese Academy of Sciences occupies the most central position in the institutional collaboration network. Research hotspots converge around four core themes: catalytic material innovation, reaction kinetics and mechanism exploration, pollutant mineralization performance, and electro-Fenton systems, reflecting the field's consistent pursuit of high degradation efficiency and in-depth mechanistic understanding. Noticeable disparities exist among countries in research priorities, keyword preferences and topic focus, which collectively constitute a diversified and complementary global research pattern.

Electrocatalytic technology has shown strong promise for water pollution control, capable of degrading a wide range of organic and inorganic pollutants with straightforward operation and high controllability for practical deployment. Ongoing advances in materials science and novel catalyst design will further improve the treatment efficiency and cost-effectiveness of electrocatalytic systems, supporting their scaling up for industrial applications.

This bibliometric analysis provides a holistic overview of the state of the art and offers quantitative support for future research planning. Going forward, the field should

prioritize the development of low-cost durable electrode materials, multi-technology process coupling, toxic by-product control, and intelligent process optimization. Strengthened cross-disciplinary integration and deeper international collaboration will help address shared bottlenecks including high energy consumption and scale-up challenges, and accelerate the translation of electrocatalytic water treatment technologies from laboratory research to real-world engineering applications.

Author contributions

Submitting author: Yaozu Xu; Yaozu Xu: Writing—Original Draft, Visualization, Software, Methodology, Data curation; Mengyao Li: Investigation, Formal analysis; Rupeng Liu: Writing—review & editing, Funding acquisition; Cuizhen Sun: Methodology, Writing—review & editing, Software, Conceptualization; Linxu Xu: Writing—review & editing, Validation; Feiyong Chen: Writing—review & editing, Validation.

Funding

This work was funded by Shandong Top Talent Special Foundation (No.0031504).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study did not involve human participants or animals. Therefore, ethical approval was not required.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 15 April 2026 / Accepted: 5 August 2026



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